

# A Local–Geometry Changepoint Aligned with the Onset of Portuguese Grammatical Competence in a Small Language Model

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Apart Research Sprint — June 2026

June 21, 2026

## Abstract

We ask whether a quantity from singular learning theory (SLT) — the *local learning coefficient* (LLC,  $\hat{\lambda}$ ), estimated by stochastic-gradient Langevin dynamics (SGLD) — changes in a way that is *aligned with a behavioral developmental transition* as a small English-trained language model acquires a second language. We adapt **TinyStories-8M** to Portuguese by full-parameter continued pretraining and track, at every checkpoint, both behavior (validation bits-per-byte and a 538-item grammatical minimal-pair benchmark) and the LLC. The model’s modeling loss falls smoothly, but grammatical competence emerges *abruptly* (minimal-pair accuracy chance  $\rightarrow$  89%, with the preference margin flipping sign at  $\approx$  2.5M tokens). Over exactly this window the LLC *rises steeply* (+52  $\rightarrow$  +87) and then plateaus — an aligned-changepoint signature. Two controls confirm specificity: a token-shuffled-Portuguese model (no word order to learn) and a matched-English model (generic adaptation) *do not* reproduce the rise-then-plateau, and the structured model is cleanly separated from both at 100M tokens (+87 vs. +28 vs. +70). The trajectory shape is robust to the SGLD localization strength. We frame this, following standard caution, as a changepoint in an SLT-derived local geometric estimate aligned with a behavioral transition — not a formal phase transition — and we are explicit about the present single-seed scope (a second-seed replication is in progress).

## 1 Introduction

Developmental interpretability asks whether internal, structural quantities of a neural network change at the moments its behavior changes. Singular learning theory provides one such quantity: the *local learning coefficient* (LLC, also called the local RLCT), a measure of the effective dimensionality / degeneracy of the loss landscape around a parameter [1, 2]. If the geometry reorganizes precisely as a capability appears, the LLC should exhibit a changepoint aligned with the behavioral one.

We study this in a deliberately small, controllable setting: second-language acquisition by continued pretraining. We take an English-trained model (**TinyStories-8M**) and adapt it to Portuguese, measuring behavior and the LLC along the whole trajectory. **The model learning Portuguese is not the result — it is the scaffolding.** The contribution is the *LLC $\leftrightarrow$ behavior alignment*, established together with the controls that make it specific. Our findings:

- Behavior shows a clean grammatical transition at  $\approx$  1.5–2.5M tokens even though bits-per-byte falls smoothly (Section 3.1).

- The LLC rises steeply through that window and then plateaus — a changepoint aligned with the behavioral one (Section 3.2).
- Two controls (shuffled-PT, matched-English) fail to reproduce the signature, so it is specific to genuine language structure being acquired, not to training length or generic adaptation (Section 3.3).
- The trajectory shape is robust to the SGLD localization hyperparameter (Section 3.4).

## 2 Setup

**Model and training.** We adapt roneneldan/TinyStories-8M by full-parameter, FP32 next-token continued pretraining (AdamW, learning rate  $10^{-4}$ , 3% warmup, cosine decay, gradient clipping 1.0, weight decay 0.01, batch size 64, sequence length 128) on a Portuguese Wikipedia corpus, to 100M tokens, with 17 log-spaced checkpoints. The recipe was fixed by a short “autoresearch” search after an initial under-trained run; we then froze it across all conditions.

**Conditions.** Following the minimal design for this question we run (i) *structured Portuguese* (the condition of interest), and two controls: (ii) *token-shuffled Portuguese* — the same tokens with within-sequence order destroyed, so there is no grammar to acquire — and (iii) *matched English* — the identical protocol on English, controlling for “any continued pretraining raises complexity.”

**Behavioral metrics (every checkpoint).** Portuguese validation bits-per-byte (BPB), English-retention BPB, and a frozen, hashed **538-item templated Portuguese grammatical minimal-pair benchmark** (10 agreement phenomena). Accuracy is the fraction of pairs for which the model assigns higher log-probability to the grammatical sentence; we also report the mean log-prob margin.

**LLC estimation.** We estimate the LLC by SGLD [2, 3] using the devinterp library, with one globally fixed sampler configuration across each trajectory (3 chains, learning rate  $10^{-5}$ ,  $n_\beta=10$ , localization 100, batch 64, 200 burn-in + 100 draws). With  $L$  the loss and  $L_0$  the loss at the checkpoint center,

$$\hat{\lambda} = n_\beta (\mathbb{E}_{\text{SGLD}}[L] - L_0), \quad (1)$$

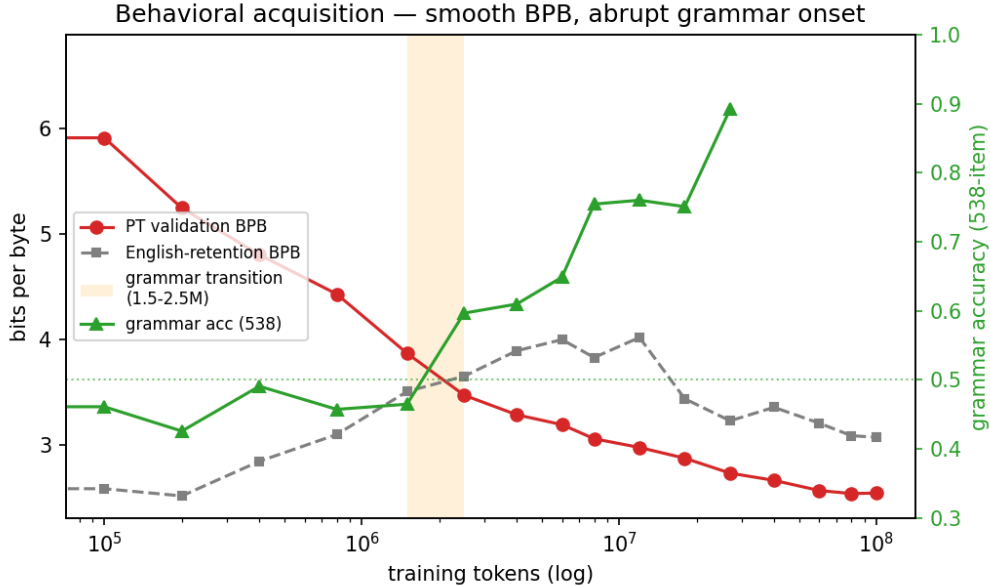
which is positive at a true local minimum of the measured loss. **Crucially, each condition is measured against a reference set drawn from its own training distribution** (full-length, non-padded chunks):  $\hat{\lambda}$  is only valid at a minimum of the loss the model actually minimized, so using one condition’s data to score another would yield meaningless estimates.

**A measurement note (why the controls matter twice).** An early version of our LLC pipeline built its reference from short sentences padded to the sequence length without masking, so  $\approx 90\%$  of every scored sequence was unmasked padding. This produced *negative*  $\hat{\lambda}$  everywhere — the checkpoint is not a minimum of *that* (padded) loss. The fix was to score non-padded chunks of the actual training distribution. We flag this because it is the central methodological lesson: an LLC estimate is only meaningful relative to a loss the model truly minimized, and one must read the *value, sign and chain agreement*, never a binary “accepted” label.

### 3 Results

#### 3.1 Behavior: smooth loss, abrupt grammar

The model genuinely acquires Portuguese: PT BPB falls monotonically  $6.69 \rightarrow 2.54$  with no kink (Figure 1). Grammatical competence, by contrast, emerges abruptly: minimal-pair accuracy sits at chance through 1.5M tokens, the mean margin flips negative $\rightarrow$ positive between 1.5M and 2.5M, and accuracy then climbs to 89.2% by 27M (Table 1). The abruptness lives in the discrete grammatical behavior, not in the modeling-loss curve.



**Figure 1:** Behavioral acquisition (structured PT, seed A). PT BPB (red) falls smoothly; the 538-item grammar accuracy (green, right axis) is flat at chance then rises sharply through the shaded 1.5–2.5M window. English-retention BPB (grey) degrades then partly recovers.

| tokens      | PT BPB | EN BPB | grammar acc. | grammar margin |
|-------------|--------|--------|--------------|----------------|
| 0           | 6.686  | 2.714  | 0.446        | −0.277         |
| 1.5M        | 3.869  | 3.506  | 0.465        | −0.394         |
| <b>2.5M</b> | 3.471  | 3.651  | <b>0.597</b> | <b>+0.461</b>  |
| 8M          | 3.057  | 3.829  | 0.755        | +1.540         |
| 27M         | 2.731  | 3.228  | 0.892        | +2.403         |
| 100M        | 2.541  | 3.071  | —            | —              |

**Table 1:** Behavioral trajectory (structured PT, seed A). The grammar margin crosses zero at  $\approx 2.5$ M tokens. Grammar was scored to 27M (the transition is well before); BPB to 100M.

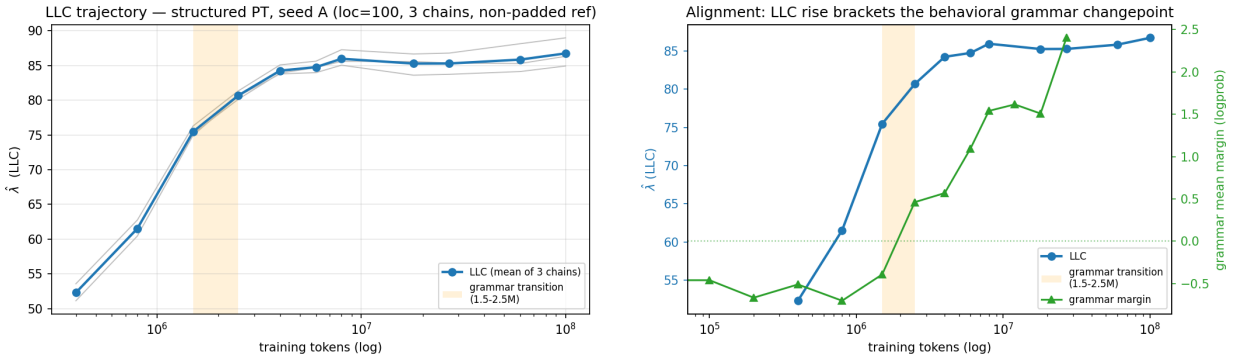
#### 3.2 The LLC trajectory: a rise-then-plateau changepoint

The LLC is positive at every checkpoint, with tight chain agreement and no rejected chains. It rises  $\approx 34$  points over 400k $\rightarrow$ 4M (steepest through 400k–2.5M) and is then flat within noise ( $\approx 84$ –87) for the remaining 25 $\times$  of training (Figure 2, left; Table 2). The steep rise *brackets* the behavioral

grammar transition: the geometry reorganizes through  $\approx 400\text{k}$ – $2.5\text{M}$ , the grammar margin crosses zero at  $\approx 2.5\text{M}$ , and both then settle (Figure 2, right). Read carefully, the LLC rise slightly *leads* the behavioral expression — local complexity increases first, grammatical behavior follows and continues maturing after  $\hat{\lambda}$  has plateaued.

| tokens                   | 400k  | 800k  | 1.5M  | 2.5M  | 8M    | 100M  |
|--------------------------|-------|-------|-------|-------|-------|-------|
| $\hat{\lambda}$ (seed A) | +52.3 | +61.5 | +75.5 | +80.7 | +86.0 | +86.7 |

**Table 2:** Primary LLC trajectory (structured PT, seed A; loc 100, 3 chains). Full 11-checkpoint trajectory in Figure 2.



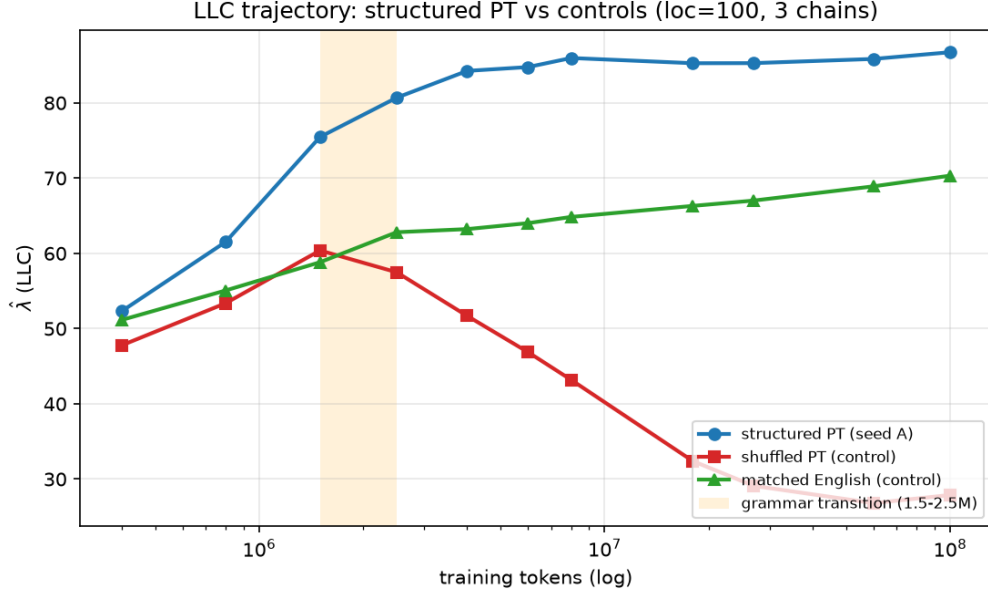
**Figure 2: Left:** the LLC trajectory (structured PT, seed A) rises steeply then plateaus; thin grey lines are individual chains. **Right:** alignment — the LLC rise (blue) brackets the grammar-margin sign change (green) in the shaded 1.5–2.5M window.

### 3.3 Controls: the changepoint is specific

Neither control reproduces the rise-then-plateau (Figure 3, Table 3). The token-shuffled model rises only weakly to  $\approx 60$  and then *declines* to  $\approx 28$  as it settles into low-complexity position-independent statistics. The matched-English model genuinely adapts (complexity rises monotonically  $51 \rightarrow 70$ ) but *smoothly and to a markedly lower level*, with no steep transition and no high plateau — the key control showing the signature is not merely “continued pretraining raises the LLC.” At 100M the three conditions are cleanly separated, and only structured Portuguese has both the steep grammar-window rise and the high plateau. The geometric changepoint is therefore specific to genuine Portuguese structure being acquired, not to training length, token statistics, or generic adaptation.

| tokens | structured PT | shuffled PT  | matched EN |
|--------|---------------|--------------|------------|
| 1.5M   | +75.5         | +60.4 (peak) | +58.8      |
| 8M     | +86.0         | +43.2        | +64.8      |
| 100M   | +86.7         | +27.9        | +70.3      |

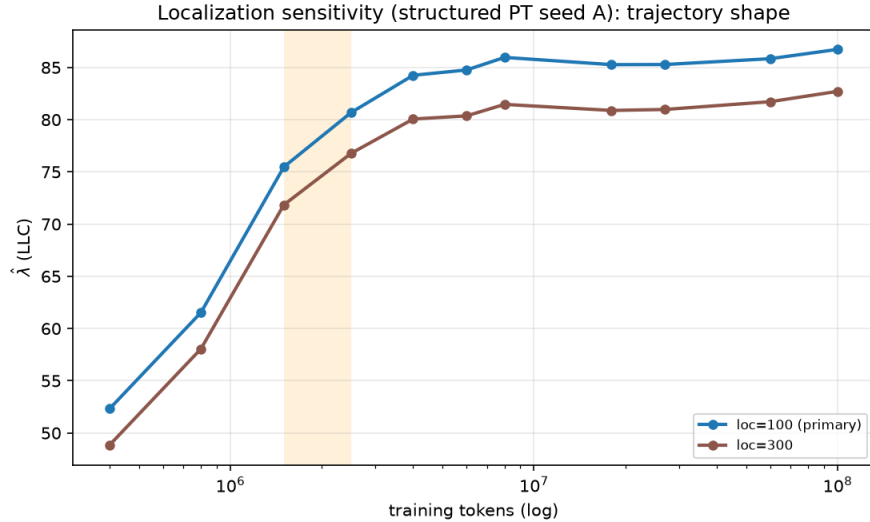
**Table 3:** LLC at shared checkpoints across the three conditions (loc 100, 3 chains, condition-matched references).



**Figure 3:** Three-condition contrast. Structured PT (blue) rises to a high plateau; shuffled PT (red) rises then declines; matched English (green) rises smoothly to a lower level. The shaded band marks the behavioral grammar transition.

### 3.4 Robustness to the localization hyperparameter

The SGLD localization  $\gamma$  sets the prior strength around the checkpoint, so it shifts the absolute level of  $\hat{\lambda}$ . At  $\gamma = 300$  the structured-PT trajectory sits  $\approx 4$  units below  $\gamma = 100$  everywhere but is *identical in shape* — same steep rise through 1.5–2.5M, same plateau (Figure 4). The changepoint is thus a property of the trajectory, not of the localization choice.



**Figure 4:** Localization sensitivity (structured PT, seed A):  $\gamma = 100$  vs.  $\gamma = 300$  are near-parallel — the rise-then-plateau shape is preserved.

## 4 Discussion

Across an identical architecture, optimizer, and token budget, three conditions produce three qualitatively different LLC trajectories, and only the one with real grammatical structure to acquire shows a steep rise bracketing the moment grammatical behavior appears. A geometric changepoint and a behavioral changepoint coincide where the modeling-loss curve shows nothing. Following standard caution we call this *a changepoint in an SLT-derived local geometric estimate aligned with a behavioral transition*, and we do not claim a formal SLT phase transition, nor grokking (training and held-out performance improve together). The slight *lead* of the LLC over behavior is suggestive of the geometry reorganizing before the capability is fully expressed, but we do not over-read a single trajectory.

## 5 Limitations

- **Single Portuguese seed.** The controls establish specificity but vary data *content*, not the random seed; they do not rule out that the exact trajectory is idiosyncratic to one initialization / data order. A genuine second-seed replication (independent data ordering) is *in progress* and will be folded in.
- **No formal changepoint test.** “The rise brackets the transition” and “the lead” are read from the trajectories, not from a fitted changepoint model with a significance statement.
- **Localization sweep is two points** ( $\gamma \in \{100, 300\}$ ); shape-invariance over a wider range would strengthen the robustness claim.
- **Templated benchmark.** The 538-item grammar set is generated from agreement templates, not a validated broad-coverage suite.

## 6 Conclusion

In a small, fully controlled language-acquisition setting, an SLT-derived local geometric quantity exhibits a changepoint aligned with the behavioral onset of grammatical competence, shown specific by two controls and robust to the SGLD localization. Pending a second-seed replication, this is, to our knowledge, a clean demonstration of an LLC $\leftrightarrow$ behavior alignment during second-language acquisition.

## References

- [1] S. Watanabe. *Algebraic Geometry and Statistical Learning Theory*. Cambridge University Press, 2009.
- [2] E. Lau, Z. Furman, G. Wei, D. Murfet, S. Wei. *Quantifying degeneracy in singular models via the learning coefficient*. 2023/2024 (devinterp).
- [3] J. Hoogland, G. Wang, M. Farrugia-Roberts, L. Carroll, S. Wei, D. Murfet. *The Developmental Landscape of In-Context Learning*. 2024.
- [4] Z. Chen et al. *Dynamical versus Bayesian Phase Transitions in a Toy Model of Superposition*. 2023.